

A Sinusoidal Hull-White Model for Interest Rate Dynamics: Capturing Long-Term Periodicity in U.S. Treasury Yields

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Why Extending the Hull-White Model?

- **Motivation of the Paper**

- The Hull-White model is a standard tool for interest rate modelling, but assumes constant mean reversion, which limits the model's ability to capture long-horizon dynamics.
- Empirical evidence suggests that interest rates exhibit long-term cyclical behavior.
- The paper proposes a simple extension allowing for smooth, time-varying mean reversion within the Hull-White framework.

- **Applications**

- Bond pricing and interest rate derivatives: improved modelling of long-dated instruments.
- Risk management: more accurate assessment of interest rate risk, supporting stress testing and scenario analysis.
- Monetary policy analysis: insights into long-term economic cycles, informing the design of forward guidance strategies.

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Positioning in the Literature

- **Hull-White Model**

- Hull and White (1990): one-factor short-rate model with constant mean reversion.
- Widely used due to analytical tractability and calibration flexibility.

- **Empirical Evidence on Cyclicalities**

- Campbell, Lo and MacKinlay (1997): interest rates exhibit long-term economic cycles.
- Bauer and Rudebusch (2018): Treasury yields show persistent cyclical behavior linked to macroeconomic forces.

- **Extensions of Hull-White**

- Focus mainly on additional factors or stochastic volatility (e.g. Heath-Jarrow-Morton, 1992; Duffie and Kan, 2000).
- Periodic dynamics are rarely modeled explicitly.

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Short-Rate Dynamics

The short-rate process under the Hull-White model is given by:

$$dr_t = \kappa(\theta - r_t) dt + \sigma dW_t$$

Here:

- r_t is the short rate at time t ,
- κ is the constant mean-reversion speed,
- θ is the long-term mean rate,
- σ is the volatility,
- dW_t is a standard Wiener process under the risk-neutral measure.

Price of ZCB and IBP

Under the risk-neutral measure $\mathbb{Q}$, the discounted price of a ZCB must be a martingale. Hence, we consider the product $D_t P(t, r_t)$, where the discount factor is

$$D_t = \exp\left(-\int_0^t r_u du\right)$$

Applying Itô's integration by parts,

$$d(D_t P(t, r_t)) = D_t dP(t, r_t) + P(t, r_t) dD_t + d\langle D, P(\cdot, r) \rangle_t$$

Since D_t has no diffusion term (it is of finite variation), we have

$$d\langle D, P(\cdot, r) \rangle_t = 0$$

In addition, trivially,

$$dD_t = -r_t D_t dt$$

Itô and Feynman-Kac

Applying Itô's formula to $P(t, r_t)$, we obtain its dynamics:

$$dP(t, r_t) = \left(\frac{\partial P}{\partial t} + \kappa(\theta - r_t) \frac{\partial P}{\partial r} + \frac{1}{2} \sigma^2 \frac{\partial^2 P}{\partial r^2} \right) dt + \sigma \frac{\partial P}{\partial r} dW_t$$

Substituting into the product dynamics,

$$\begin{aligned} d(D_t P(t, r_t)) &= D_t \left(\frac{\partial P}{\partial t} + \kappa(\theta - r_t) \frac{\partial P}{\partial r} + \frac{1}{2} \sigma^2 \frac{\partial^2 P}{\partial r^2} - r_t P \right) dt \\ &\quad + D_t \sigma \frac{\partial P}{\partial r} dW_t \end{aligned}$$

For $D_t P(t, r_t)$ to be a martingale under $\mathbb{Q}$, its drift must be zero. Thus we obtain the Feynman-Kac PDE:

$$\boxed{\frac{\partial P}{\partial t} + \kappa(\theta - r_t) \frac{\partial P}{\partial r} + \frac{1}{2} \sigma^2 \frac{\partial^2 P}{\partial r^2} - r_t P = 0, \quad P(T, \cdot) = 1}$$

Solving the PDE

We look for a solution of the form

$$P(t, T) = A(t, T) e^{-B(t, T)r_t}$$

Computing the derivatives:

$$\frac{\partial P}{\partial t} = \left(\frac{\partial A}{\partial t} - A \frac{\partial B}{\partial t} r_t \right) e^{-Br_t},$$

$$\frac{\partial P}{\partial r} = -AB e^{-Br_t}, \quad \frac{\partial^2 P}{\partial r^2} = AB^2 e^{-Br_t}$$

Substituting into the PDE and simplifying the exponential term:

$$\frac{\partial A}{\partial t} - A \frac{\partial B}{\partial t} r_t + \frac{1}{2} \sigma^2 AB^2 - \kappa \theta AB + \kappa AB r_t - r_t A = 0$$

Solving the PDE

The coefficient of r_t yields the following ODE:

$$\frac{\partial B(t, T)}{\partial t} = \kappa B(t, T) - 1, \quad B(T, T) = 0$$

Separating variables and integrating:

$$\int \frac{dB}{\kappa B - 1} = \int dt \implies \frac{1}{\kappa} \ln |\kappa B - 1| = t + C$$

$$B(t, T) = \frac{1}{\kappa} (1 - Ce^{\kappa t})$$

Applying the terminal condition $B(T, T) = 0$:

$$B(t, T) = \frac{1}{\kappa} (1 - e^{-\kappa(T-t)})$$

Solving the PDE

From the constant term we obtain the ODE for $A(t, T)$:

$$\frac{\partial A}{\partial t} = A \left(\kappa \theta B - \frac{1}{2} \sigma^2 B^2 \right), \quad A(T, T) = 1$$

$$\ln A(t, T) = \int_t^T \left(\kappa \theta B(s, T) - \frac{1}{2} \sigma^2 B(s, T)^2 \right) ds + C$$

Applying the terminal condition $A(T, T) = 1$:

$$A(t, T) = \exp \left(\int_t^T \left(\kappa \theta B(s, T) - \frac{1}{2} \sigma^2 B(s, T)^2 \right) ds \right)$$

Substituting $B(s, T)$ and evaluating the integral:

$$A(t, T) = \exp \left(\left(\theta - \frac{\sigma^2}{2\kappa^2} \right) (B(t, T) - (T - t)) - \frac{\sigma^2}{4\kappa} B(t, T)^2 \right)$$

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Short-Rate Dynamics and Feynman-Kac

Under the sinusoidal Hull-White model, the short-rate dynamics are given by:

$$dr_t = [\kappa_0 + A \sin(\omega t)](\theta - r_t) dt + \sigma dW_t$$

Here:

- κ_0 is the baseline mean-reversion speed,
- A is the amplitude of the sinusoidal component,
- ω is the angular frequency (in radians per day).

The corresponding Feynman-Kac PDE for the ZCB price $P(t, T)$ is:

$$\frac{\partial P}{\partial t} + \frac{1}{2}\sigma^2 \frac{\partial^2 P}{\partial r^2} + [\kappa_0 + A \sin(\omega t)](\theta - r_t) \frac{\partial P}{\partial r} - r_t P = 0$$

Solving the PDE

We attempt again a solution of the form

$$P(t, T) = A(t, T) e^{-B(t, T)r_t}$$

After computing the derivatives:

$$\frac{\partial A}{\partial t} - A \frac{\partial B}{\partial t} r_t + \frac{1}{2} \sigma^2 A B^2 - [\kappa_0 + A \sin(\omega t)] \theta A B$$

$$+ [\kappa_0 + A \sin(\omega t)] A B r_t - r_t A = 0$$

ODE from the r_t -term:

$$\frac{\partial B(t, T)}{\partial t} = [\kappa_0 + A \sin(\omega t)] B(t, T) - 1, \quad B(T, T) = 0$$

Solving the PDE?

Using the integrating factor method:

$$\frac{d}{dt} \left(B(t, T) e^{-\int^t [\kappa_0 + A \sin(\omega u)] du} \right) = - e^{-\int^t [\kappa_0 + A \sin(\omega u)] du}$$

Hence, the solution for $B(t, T)$ can be written as:

$$B(t, T) = - e^{(\kappa_0 t - \frac{A}{\omega} \cos(\omega t))} \int_t^T e^{-(\kappa_0 s - \frac{A}{\omega} \cos(\omega s))} ds$$

Even worse, the equation for $A(t, T)$:

$$\frac{\partial A}{\partial t} = A \left([\kappa_0 + A \sin(\omega t)] \theta B - \frac{1}{2} \sigma^2 B^2 \right)$$

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Calibration Procedure

The calibration problem is formulated as the following optimization:

$$\min_{\theta} \sum_{T_i} (P_{\text{model}}(0, T_i; \theta) - P_{\text{observed}}(T_i))^2$$

Here, θ denotes the parameter vector:

- $\theta = (\kappa, \theta, \sigma)$ for the standard Hull-White model,
- $\theta = (\kappa_0, A, \omega, \theta, \sigma)$ for the sinusoidal Hull-White model.

⇒ **Nelder-Mead Simplex Algorithm**: a derivative-free optimization method, chosen for its robustness in handling nonlinear and potentially non-convex objective functions.

Monte Carlo Simulation for Bond Pricing

Euler-Maruyama scheme to simulate the short-rate paths:

$$r_{t+\Delta t} = r_t + [\kappa_0 + A \sin(\omega t)](\theta - r_t)\Delta t + \sigma\sqrt{\Delta t}\varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, 1)$$

Under the risk-neutral measure, the ZCB price satisfies:

$$P(0, T) = \mathbb{E}\left[\exp\left(-\int_0^T r_s ds\right)\right]$$

Hence, the Monte Carlo estimator for the bond price is (approximation of the integral via a Riemann sum):

$$P(0, T) \approx \frac{1}{N} \sum_{i=1}^N \exp\left(-\sum_{j=0}^{T/\Delta t-1} r_{j,i} \Delta t\right)$$

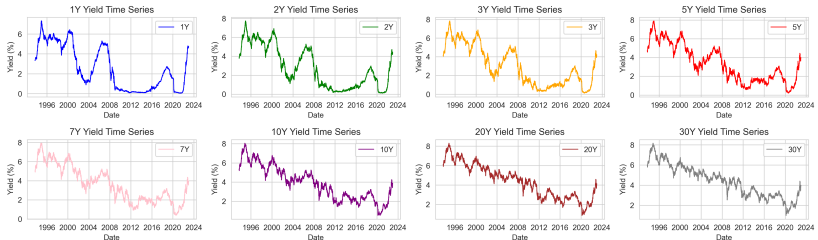
where $r_{j,i}$ denotes the short rate at the j -th time step along the i -th simulated path.

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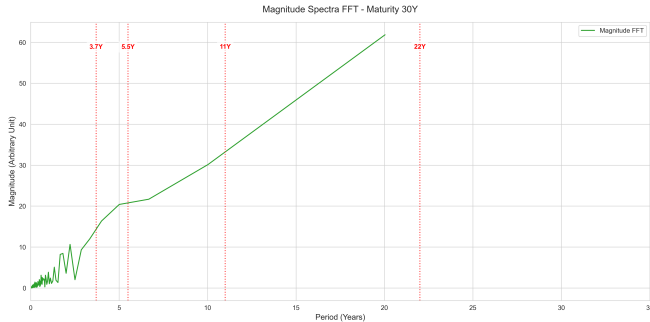
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Data

U.S. Treasury Yields (1990-2022)



Periodicity Analysis



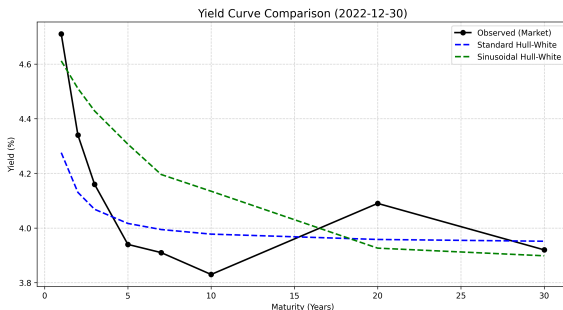
$$\Rightarrow \omega = \frac{2\pi}{365 \times 20.03} = 0.000859 \approx 0.00078 \text{ in the paper.}$$

Validation of the Monte Carlo Pricer

Maturity	Analytical Price (HW)	MC Price (HW)	Diff (Error)
1	0.9581	0.9578	-0.00031
2	0.9207	0.9205	-0.00019
3	0.8851	0.8858	0.00072
5	0.8180	0.8177	-0.00036
7	0.7561	0.7555	-0.00061
10	0.6718	0.6723	0.00050
20	0.4531	0.4533	0.00023
30	0.3056	0.3052	-0.00036

Calibration Results

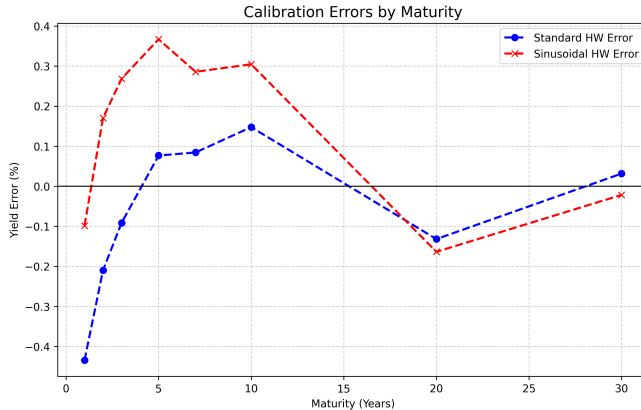
- **Standard Hull-White:** $\kappa = 1.99$, $\theta = 0.0395$, $\sigma = 0.0301$, $RMSE = 6.28 \times 10^{-3}$
- **Sinusoidal Hull-White:** $\kappa_0 = 0.265$, $A = 0.214$, $\omega = 0.000859$, $\theta = 0.0389$, $\sigma = 0.0090$, $RMSE = 8.85 \times 10^{-3}$



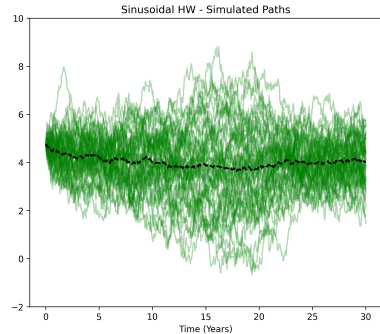
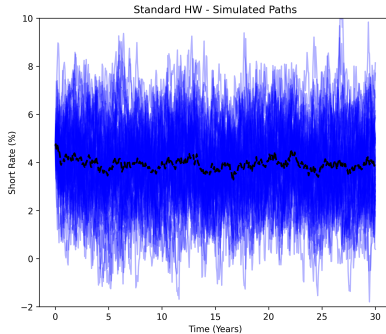
Calibration Results

Maturity	Market Yield (%)	HW Yield (%)	Sin HW Yield (%)
1	4.71	4.28	4.61
2	4.34	4.13	4.51
3	4.16	4.07	4.43
5	3.94	4.02	4.31
7	3.91	3.99	4.20
10	3.83	3.98	4.13
20	4.09	3.96	3.93
30	3.92	3.95	3.90

Calibration Errors



Simulation of Short-Rate Trajectories



Conclusions

- **Minor differences** with respect to the reference paper can be attributed to alternative data-cleaning procedures, Monte Carlo randomness in bond pricing, and the absence of an explicit calibration date in the original study.
- Calibration results show that the **sinusoidal Hull-White model improves the fit for long maturities**, particularly at the 30-year tenor, while the standard model performs better at intermediate maturities.

Conclusions

- However, the model entails increased computational costs and shares some of the **limitations** of the standard Hull-White framework, including the possibility of negative interest rates.
- **Possible extensions** include allowing for stochastic volatility, incorporating multi-factor dynamics, and exploring alternative functional forms for the time-varying mean-reversion speed to further improve the model's flexibility and performance.

Thank you for your attention!
Do you have any questions?